

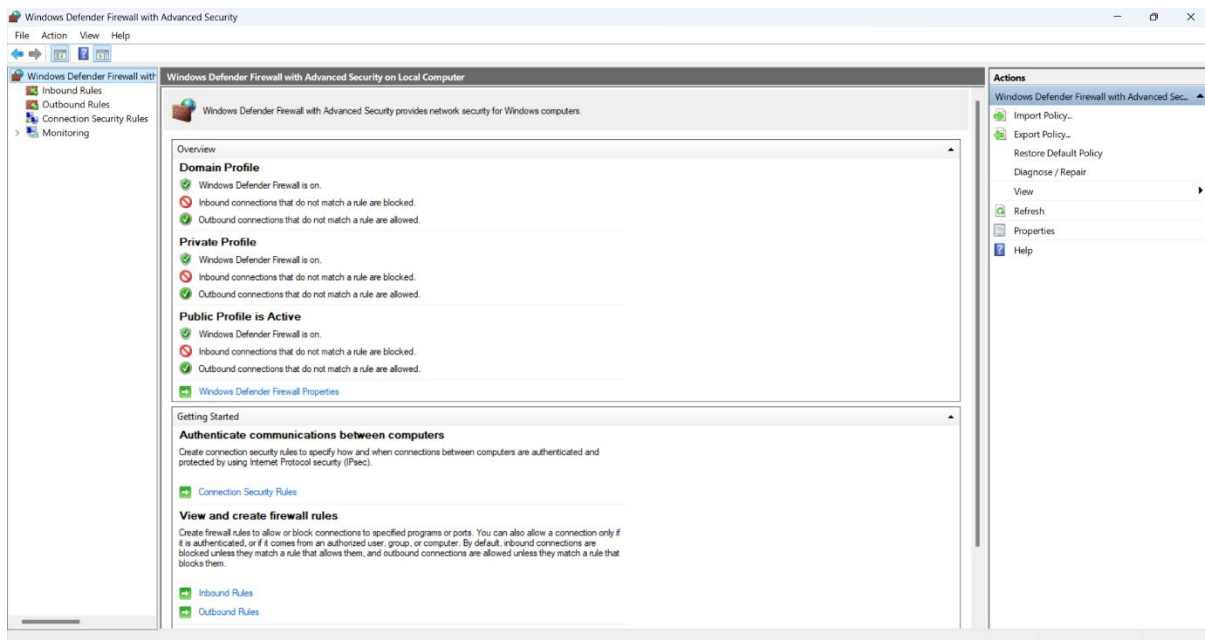
Secure networks with firewalls NAT, port filtering, packet filtering, and other methods

Q1. Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.

Ans: Steps to Block Incoming Traffic on Port 8080

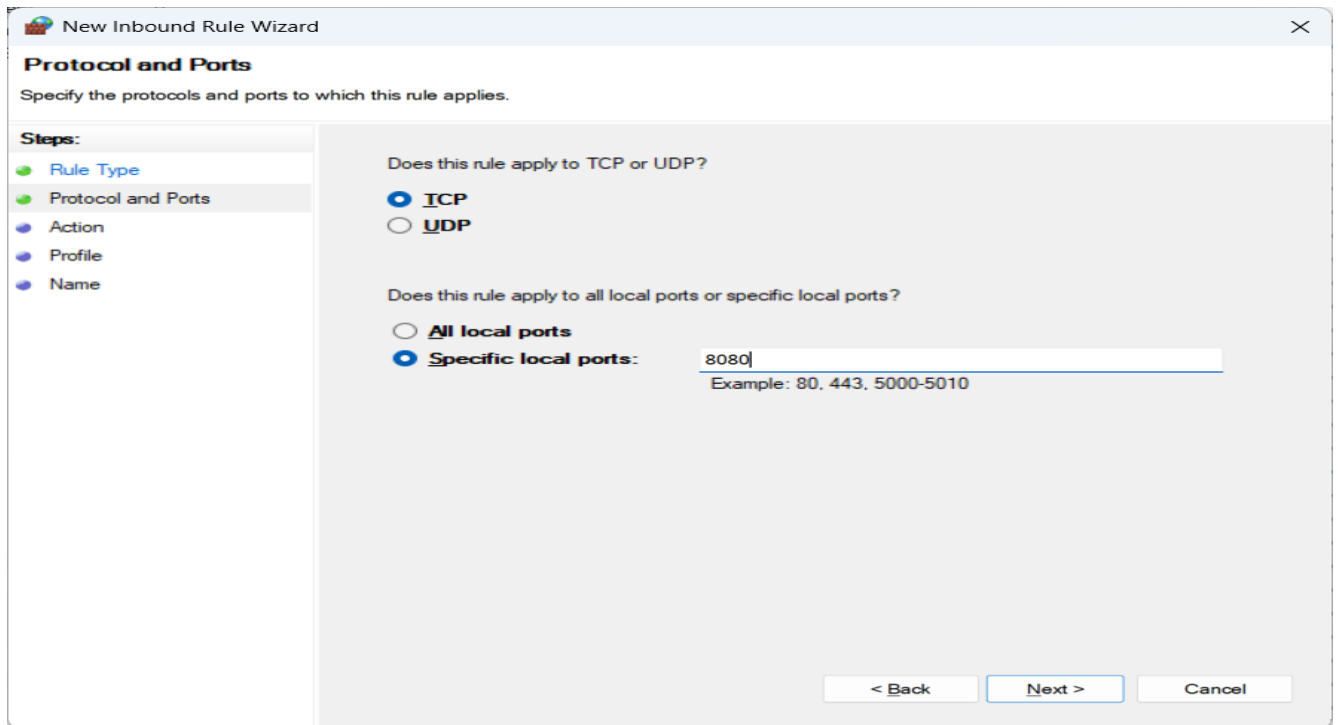
1. Open **Windows Defender Firewall with Advanced Security** by running **wf.msc**.

Figure 1: Windows Defender Firewall with Advanced Security



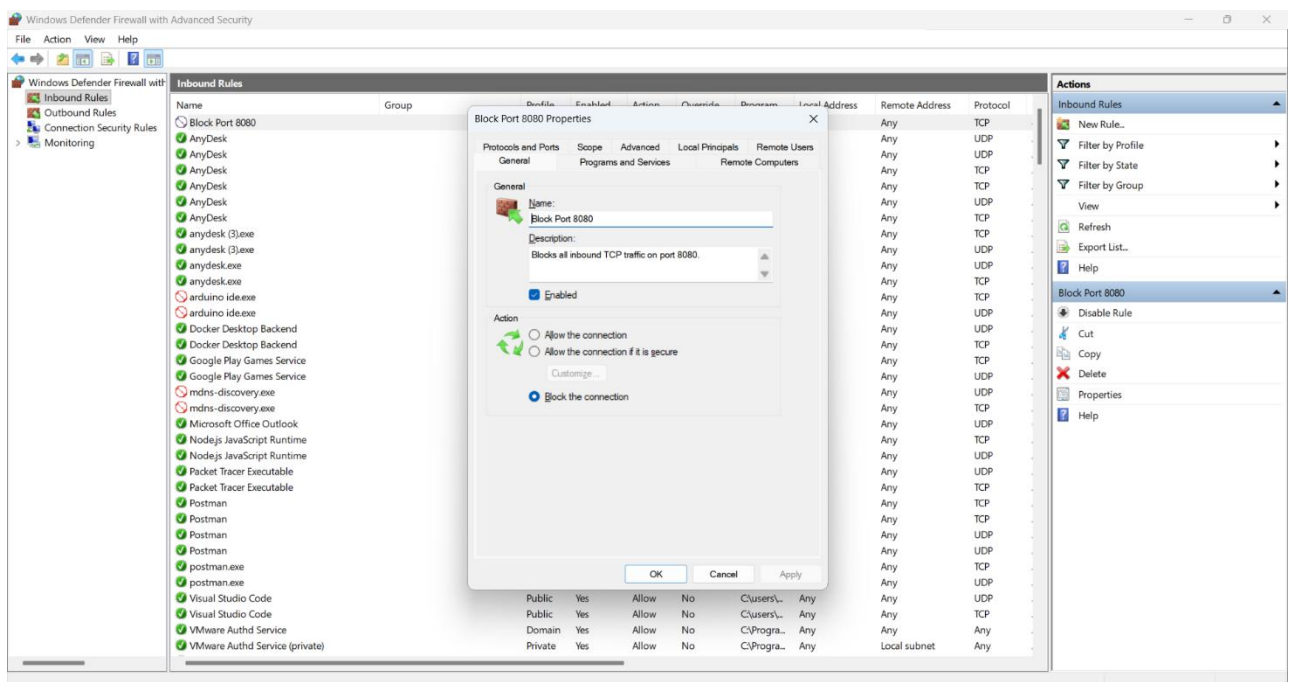
2. Click **Inbound Rules**.
3. Select **New Rule**.
4. Choose **Port** and click **Next**.
5. Select **TCP** and enter **8080** as the specific local port.

Figure 2: Creating a New Inbound Firewall Rule for TCP Port 8080



6. Select **Block the connection**.
7. Apply the rule to **Domain, Private, and Public** profiles.
8. Enter the rule name **Block Port 8080**.
9. Click **Finish** to create the firewall rule.

Figure 3: Successfully Created Inbound Rule to Block TCP Port 8080

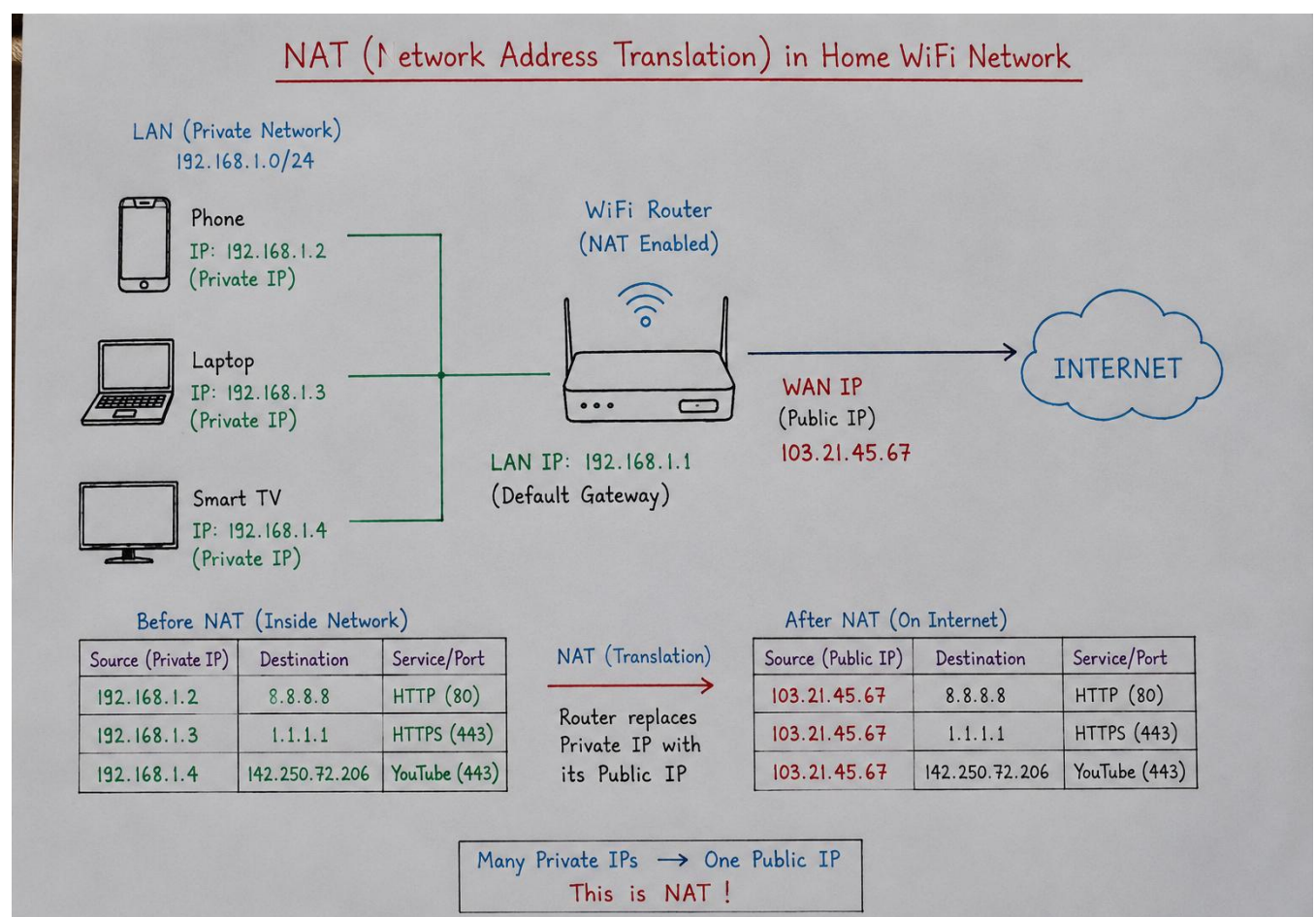


Q2. Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.
Hint: Use any drawing tool or even hand-draw and click a photo.

Ans: NAT:

Network Address Translation (NAT) is a technique used by a router to translate multiple private IP addresses into a single public IP address. This allows multiple devices to access the Internet using one public IP while hiding the internal private IP addresses.

Figure 1: NAT (Network Address Translation) in a Home Wi-Fi Network



Working of NAT

- The router assigns private IP addresses to devices on the local network.
- When devices access the Internet, the router replaces their private IP addresses with a single public IP address.

- The router keeps track of each connection and forwards the response to the correct device.

Result

NAT allows multiple devices to share one public IP address, conserves IPv4 addresses, and improves network security by hiding private IP addresses.

Q3. Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

Ans: Steps to Configure Port Filtering

1. Open the router's administration panel or a router simulator.
2. Navigate to Security, Firewall, or Access Control settings.
3. Open the Port Filtering or Outbound Filtering section.
4. Create a new filtering rule.
5. Configure the rule with the following settings:
 - **Rule Name:** Block SMTP
 - **Traffic Direction:** Outgoing
 - **Protocol:** TCP
 - **Destination Port:** 25 (SMTP)
 - **Action:** Block / Deny
6. Enable the rule and save the configuration.
7. Apply the changes. If required, restart the router to activate the new rule.

Figure 1: Router Simulator – Port Filtering Configuration for Blocking Outgoing SMTP (Port 25)

Filter

Filter Type

Filter Type Selection : IP / MAC Filter

IP / MAC Filter Set Editing

IP / MAC Filter Set Index : 1

Interface : PVC0

Direction : Outgoing

IP / MAC Filter Rule Editing

IP / MAC Filter Rule Index : 1

Rule Type : IP

Active : ☒ Yes ☐ No

Source IP Address : 0.0.0.0 (0.0.0.0 means Don't care)

Subnet Mask : 0.0.0.0

Port Number : 0 (0 means Don't care)

Destination IP Address : 0.0.0.0 (0.0.0.0 means Don't care)

Subnet Mask : 0.0.0.0

Port Number : 25 (0 means Don't care)

Protocol : TCP

Rule Unmatched : Forward

IP / MAC Filter Listing

#	Active	Src Address/Mask	Dest IP/Mask	Src Port	Dest Port	Protocol	Unmatched
1	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-
6	-	-	-	-	-	-	-

SAVE DELETE CANCEL

Explanation

Blocking outgoing traffic on port 25 (SMTP) prevents unauthorized devices or malware from sending spam emails directly to external mail servers.

This helps protect the network from spam abuse and reduces the risk of the public IP address being blacklisted.

Q4. Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53. Hint: Focus on source/destination IPs and ports in the rule set.

Ans:

Packet Filtering Rule Set (Provided by Trainer)

Rule	Action
Incoming Port 80	Block
Incoming Port 22	Block

Sample Packet	Status	Reason
HTTP request from 192.168.1.5 to 172.217.166.78:80	Allowed	This is an outgoing HTTP request. The firewall rule blocks only incoming traffic on port 80.
SSH request from 10.0.0.10 to 34.201.23.1:22	Allowed	This is an outgoing SSH request. The firewall rule blocks only incoming traffic on port 22.
DNS query from 192.168.1.7 to 8.8.8.8:53	Allowed	This is an outgoing DNS request using port 53, which is not blocked by the firewall rules.

Q5. Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

Ans:

1. Blocking Unauthorized Access

- Instagram and Zomato use firewalls to ensure that only authorized users and trusted services can access their servers.
- Any unauthorized or suspicious connection attempts are automatically blocked to protect their network.

2. Filtering Malicious Network Traffic (Requests)

- Packet filtering examines incoming and outgoing network packets based on IP addresses, protocols, and port numbers.
- Harmful or suspicious packets are blocked before they reach the application servers, reducing the risk of cyberattacks.

3. Protecting Against DDoS and Bot Attacks

- Firewalls and packet filtering help detect and filter excessive traffic generated by bots or Distributed Denial-of-Service (DDoS) attacks.
 - This prevents attackers from overwhelming the servers and ensures that the applications remain available for legitimate users.
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Conclusion

Instagram and Zomato use firewalls and packet filtering to block unauthorized access, filter malicious network traffic, and protect their servers from attacks such as DDoS. These security measures help keep their services reliable, secure, and available for users.